

GARZE, China

WMO: 561460

Lat: 31.6191N

Lon: 99.9976E

Elev: 11135

StdP: 9.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.2	(c) 10.5	(d) -15.9	(e) 3.5	(f) 35.0	(g) -12.0	(h) 4.4	(i) 31.8	(j) 22.6	(k) 43.6	(l) 19.3	(m) 41.6	(n) 1.1	(o) 20	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.2	(c) 76.2	(d) 53.5	(e) 73.7	(f) 52.9	(g) 71.2	(h) 52.3	(i) 57.2	(j) 69.9	(k) 55.9	(l) 68.1	(m) 54.9	(n) 66.3	(o) 5.1	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
53.0	91.4	61.5	52.0	88.0	60.1	51.0	84.9	59.4	30.2	70.1	29.2	68.2	28.3	66.9	62.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.9	(b) 16.0	(c) 12.7	(e) 1.1	(f) 81.4	(g) 3.6	(h) 2.1	(i) -1.5	(j) 82.9	(k) -3.6	(l) 84.2	(m) -5.6	(n) 85.4	(o) -8.2	(p) 86.9		
(a) 19.9	(b) 16.0	(c) 12.7	(e) -0.6	(f) 59.5	(g) 3.3	(h) 1.8	(i) -3.0	(j) 60.8	(k) -4.9	(l) 61.9	(m) -6.7	(n) 62.9	(o) -9.1	(p) 64.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	28.0	32.9	38.8	45.7	52.6	57.2	59.5	59.2	55.5	46.4	36.0	28.9
	DBStd	12.46	6.13	5.37	5.19	5.02	5.47	4.60	3.88	4.03	4.85	5.61	5.57	5.16
	HDD50	2920	683	478	349	145	34	3	1	1	11	141	419	655
	HDD65	7266	1148	898	813	578	385	237	174	182	285	577	869	1120
	CDD50	1137	0	0	1	17	114	218	295	287	176	29	0	0
	CDD65	8	0	0	0	0	0	2	3	3	0	0	0	0
	CDH74	191	0	0	0	1	22	37	47	63	19	2	0	0
	CDH80	5	0	0	0	0	0	2	1	2	0	0	0	0
Wind	WSAvg	4.1	3.8	4.5	5.2	5.2	5.0	4.2	3.7	3.6	3.7	3.6	3.3	3.1
Precipitation	PrecAvg	25.0	0.2	0.2	0.6	1.2	2.8	5.2	4.6	3.8	4.4	1.5	0.3	0.2
	PrecMax	31.9	0.6	0.6	2.6	3.0	5.6	7.8	7.6	7.5	6.2	3.3	1.8	0.6
	PrecMin	19.3	0.0	0.0	0.0	0.4	0.8	3.1	2.4	0.7	0.3	0.2	0.0	0.0
	PrecStd	2.9	0.2	0.2	0.5	0.7	1.2	1.1	1.4	1.4	1.3	0.8	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	57.3	63.7	70.2	77.2	78.5	78.3	79.1	76.6	71.6	59.8	53.9
		MCWB	33.2	34.1	39.2	43.7	50.1	53.1	55.7	53.4	53.0	49.1	37.3	33.3
	2%	DB	49.2	53.1	58.5	65.3	72.4	73.7	74.9	75.5	72.7	65.6	55.7	49.2
		MCWB	30.4	32.7	36.7	41.2	47.4	53.7	55.4	53.6	52.1	46.3	35.9	31.1
	5%	DB	45.2	49.6	54.9	61.6	68.8	70.5	71.9	72.6	69.4	61.6	52.5	45.4
		MCWB	28.6	31.2	35.2	39.9	46.1	53.0	54.6	53.4	51.7	44.8	34.6	29.3
	10%	DB	41.6	46.1	51.8	58.3	65.2	67.4	69.1	69.6	66.0	57.6	49.2	41.7
		MCWB	27.1	29.9	33.9	39.0	45.4	52.1	54.2	52.9	50.9	42.8	33.2	27.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	35.2	40.6	45.8	53.1	58.2	59.1	58.1	55.8	50.6	39.6	34.0
		MCDB	53.8	53.2	58.9	64.3	68.9	71.6	72.4	70.9	70.3	66.6	52.7	51.1
	2%	WB	31.2	33.4	38.0	43.3	50.5	56.0	57.6	56.2	54.2	48.6	37.3	31.5
		MCDB	47.6	50.5	54.5	59.4	66.5	68.6	70.4	69.0	68.0	63.0	51.9	47.1
	5%	WB	29.2	31.9	36.3	41.8	48.7	54.7	56.4	55.0	52.9	46.6	35.7	29.7
		MCDB	43.9	47.5	51.4	56.5	63.5	66.6	68.4	67.3	65.7	58.1	49.6	43.8
	10%	WB	27.4	30.4	34.9	40.3	47.2	53.4	55.3	54.0	51.7	44.6	34.1	28.0
		MCDB	40.8	44.0	48.7	54.2	60.5	64.4	66.3	65.3	63.1	54.2	47.0	40.6
Mean Daily Temperature Range	5% DB	MDBR	27.8	26.2	23.8	23.0	21.8	19.0	19.2	20.3	20.6	21.9	26.6	28.2
		MCDBR	33.2	32.1	29.5	29.0	27.7	23.9	25.0	26.4	25.8	27.8	32.0	33.2
		MCWBR	18.3	16.3	13.0	11.0	9.0	8.2	8.8	8.5	8.9	10.7	14.6	17.9
	5% WB	MCDBR	32.0	29.9	25.9	24.4	23.1	20.0	20.2	20.9	22.2	23.8	28.6	31.4
MCWBR		18.0	15.3	11.5	9.6	8.5	8.0	8.0	7.5	8.2	9.6	13.3	17.3	
Clear-Sky Solar Irradiance	taub	0.224	0.257	0.310	0.328	0.313	0.302	0.307	0.307	0.279	0.251	0.225	0.211	
	taud	2.569	2.482	2.313	2.315	2.442	2.587	2.592	2.571	2.648	2.663	2.644	2.641	
	Ebn at Noon	328	322	308	305	308	310	308	308	316	322	325	329	
	Edh at Noon	25	30	39	41	36	31	31	31	28	25	23	22	
All-Sky Solar Radiation	RadAvg	1216	1464	1803	2069	2026	1786	1807	1714	1545	1374	1299	1172	
	RadStd	69	82	102	147	180	153	131	190	114	66	71	63	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.97	N/A	-3.49	N/A	-0.76	-0.68	N/A	-310	+158	+5
(47) Regional (5 neighbors)		+0.88	+1.10	-2.79	+1.39	N/A	N/A	-137	-292	+211	+13

Nomenclature: See separate page